Notes: De Giorgi, Frederiksen & Pistaferri (RES, 2020)
Consumption Network Effects

Contents

1 Big picture 3

2 Data and measurement 3
- 2.1 Administrative tax and employment data . . . 3
- 2.2 Consumption measurement from tax records . . . 4
- 2.3 Danish Expenditure Survey validation . . . 4
- 2.4 Peer network construction . . . 4

3 Models and empirical specifications 4
- 3.1 A general peer-effects Euler-equation logic . . . 4
- 3.2 Demand-system interpretation . . . 5
- 3.3 Main first-difference peer-effects regression . . . 5
- 3.4 Instrumental-variables strategy . . . 5
- 3.5 Risk-sharing test . . . 5

4 Identification and empirical design 6
- 4.1 Why identification is difficult . . . 6
- 4.2 Why spouse links matter . . . 6
- 4.3 Robustness logic . . . 6

5 Tables and figures 6
- 5.1 Table 1 and Figure 1: data quality and sample description . . . 6
- 5.2 Table 2 and Figure 2: mechanisms and network identification . . . 7
- 5.3 Table 3 and Table 4: network scale and baseline peer effects . . . 8
- 5.4 Figure 3 and Table 5: heterogeneity and correlated effects . . . 8
- 5.5 Tables 6 and 7: measurement, weighting, and the saving channel . . . 9
- 5.6 Table 8 and Figure 4: demand estimation and visible goods . . . 10
- 5.7 Tables 9 and 10: risk sharing and aggregate policy simulations . . . 10

6 Short summary 11

7 Conclusion 11
- 7.1 Core conclusion . . . 11
- 7.2 Mechanism conclusion . . . 12
- 7.3 Identification conclusion . . . 12
- 7.4 Policy implication . . . 12
- 7.5 Welfare implication . . . 12
- 7.6 Broader takeaway . . . 12

7.7 Limitations and future research 12

1 Big picture

- **Question.** Does the consumption of peers affect household consumption? If so, how large is the effect, what mechanism generates it, and what are the macro implications?
- **Empirical setting.** The paper uses Danish administrative data from 1980–1996. The data contain income, assets, employer identifiers, plant identifiers, worker characteristics, and family structure. Consumption is constructed from tax records using budget accounting.
- **Network definition.** The reference group is a workplace network. A peer is a co-worker in the same plant and similar occupation/education cell. Since both spouses work, the household is connected to two workplace networks.
- **Identification idea.** Own household consumption may move with peers because of common shocks, sorting, or reflection. The paper uses distance-3 network nodes and firm-level shocks to peers of peers as instruments for peer consumption.
- **Main result.** The preferred IV estimates imply an elasticity of household consumption with respect to average peer consumption of roughly 0.3. Separate wife and husband peer effects are about 0.30 and 0.37.
- **Scale interpretation.** A 10% increase in average peer consumption is a large aggregate shock because it means all peers increase consumption. At the random-peer level, the effect is small but non-negligible because networks are large.
- **Mechanism.** The evidence points toward *intertemporal* peer effects: households appear to adjust total spending/saving in response to peers. The paper finds little evidence that peer effects distort budget shares toward visible goods.
- **Risk-sharing test.** The paper does not find support for the hypothesis that correlated consumption is mainly due to risk sharing among co-workers.
- **Aggregate implication.** Network structure matters for fiscal multipliers. The same transfer can have different aggregate and distributional effects depending on whether it targets high-consumption, low-consumption, or random households.
- **Main takeaway.** Co-worker networks transmit consumption behavior. The effect is economically meaningful, operates mainly through total spending rather than demand composition, and makes policy multipliers depend on network connectedness.

2 Data and measurement

2.1 Administrative tax and employment data

- The core data are Danish administrative records for 1980–1996.
- The data contain income, assets, family links, employer IDs, plant IDs, sector, occupation, schooling, age, household composition, and employment status.
- The analysis focuses on married couples in which both spouses work as employees. This restriction is needed because household consumption is observed at the household level, while peer networks are formed through each spouse's workplace.
- Co-workers are identified at the plant level and weighted using similarity in occupation and education.

Interpretation: The data allow the paper to construct peer networks much more precisely than studies that define peers by geography, race, or broad demographics. The key empirical advantage is that workplace links are directly observed and can be combined with spouse links.

2.2 Consumption measurement from tax records

Consumption is not directly observed in tax records. The paper constructs spending using the household budget constraint:

$$C_{it} \approx (Y_{it} - T_{it}) - \Delta A_{it}, \tag{1}$$

where Y_{it} is income, T_{it} is taxes, and ΔA_{it} is the change in assets.

The main outcome is log adult-equivalent consumption:

$$\ln C_{it}. \tag{2}$$

Interpretation: This measure is close to total spending rather than a narrow consumption category. It is powerful because it can be constructed for a very large population panel, but it can be noisy because capital gains, capital losses, durable purchases, and asset reporting affect inferred consumption.

2.3 Danish Expenditure Survey validation

The authors match the tax-record consumption measure to the Danish Expenditure Survey for 1994–1996.

- Figure 1 compares distributions and household-level matched observations.
- The two measures have similar central mass but differ in tails.
- The matched scatterplot shows positive alignment, but substantial noise remains.

Interpretation: The validation exercise supports using tax records for large-scale panel estimation, while also motivating robustness checks that trim the consumption distribution or restrict the sample to households less likely to have capital-gain measurement problems.

2.4 Peer network construction

For household i, the paper constructs two peer-consumption variables:

$$\overline{\ln C}^{w}_{it}, \qquad \overline{\ln C}^{h}_{it}, \tag{3}$$

where w denotes the wife's co-worker network and h denotes the husband's co-worker network.

- **Distance 1:** direct co-workers of the wife or husband.
- **Distance 2:** households reached through the spouses of those co-workers.
- **Distance 3:** co-workers of those spouses, used for instruments and identification.

Interpretation: The household-level structure is crucial. Since consumption is joint within the household but workplace networks are individual, spouses create bridges between otherwise separate workplace networks.

3 Models and empirical specifications

3.1 A general peer-effects Euler-equation logic

The paper embeds peer consumption in a framework where households choose consumption over time and across goods. In a simplified form, the Euler equation becomes

$$\Delta \ln C_{it+1} = \rho^{-1}(r - \phi) + \kappa \Delta X_{it} - \Delta \ln b_{t+1}(\cdot) + \lambda \Delta \left(\frac{\overline{\ln C}_{it+1}}{a_{t+1}(\cdot)} \right) + \varepsilon_{it+1}. \tag{4}$$

Interpretation: The coefficient on peer consumption can reflect intertemporal distortions: households may borrow less, save less, or shift spending over time to keep up with the consumption path of their peers.

3.2 Demand-system interpretation

A separate mechanism is intratemporal imitation or conspicuous consumption. If peers affect demand composition, then peer consumption should shift budget shares toward visible goods:

$$s_{it}^{j} = \alpha_j + \beta_j \ln C_{it} + \theta_j \overline{\ln C}_{it} + X_{it}'\gamma_j + u_{it}^{j}, \tag{5}$$

where s_{it}^{j} is the budget share of good category j.

Interpretation: If the peer effect is about status or visibility, θ_j should be larger for goods that are easier for peers to observe. This is why the paper studies the Heffetz visibility index.

3.3 Main first-difference peer-effects regression

The central empirical specification is a first-difference model:

$$\Delta \ln C_{it} = \delta_0 + \theta_w \Delta \overline{\ln C}_{it}^{w} + \theta_h \Delta \overline{\ln C}_{it}^{h} + \gamma_w \Delta \overline{X}_{it}^{w} + \gamma_h \Delta \overline{X}_{it}^{h} + \delta_w \Delta X_{it}^{w} + \delta_h \Delta X_{it}^{h} + \varepsilon_{it}. \tag{6}$$

The average-peer version imposes $\theta_w = \theta_h = \theta$:

$$\Delta \ln C_{it} = \delta_0 + \theta \Delta \overline{\ln C}_{it} + \Gamma' \Delta Z_{it} + \varepsilon_{it}. \tag{7}$$

Interpretation: First differences remove time-invariant household heterogeneity. The peer-consumption coefficient measures whether a household changes its consumption when its co-worker networks change consumption.

3.4 Instrumental-variables strategy

The peer-consumption variable is endogenous because of reflection, sorting, and common shocks. The paper instruments peer consumption using:

- characteristics of distance-3 peers;
- firm-level shocks to distance-3 peers' employers;
- changes in firm size, public-sector status, growth, and firm type.

The exclusion restriction is that shocks to the co-workers of the spouses of one's co-workers affect own consumption only through the consumption of direct peers.

Interpretation: The instrument uses the network geometry. Friends-of-friends-of-friends are far enough to plausibly avoid direct consumption spillovers, but close enough to shift the consumption of direct peers.

3.5 Risk-sharing test

The paper compares survey consumption and tax-record consumption:

$$\ln C_{it}^{S} - \ln C_{it}^{T} = X_{it}'\pi_0 + \pi_1 \Delta \ln Y_{it} + \pi_2 \Delta \overline{\ln Y}_{it} + v_{it}. \tag{8}$$

Risk sharing predicts

$$\pi_1 < 0, \qquad \pi_2 > 0. \tag{9}$$

Interpretation: If co-workers insure each other through informal transfers, unlucky households should have higher actual spending than tax-record-based spending, and lucky peer groups should transfer resources toward them. The sign restrictions test that idea.

4 Identification and empirical design

4.1 Why identification is difficult

- **Reflection problem:** if my consumption affects peers and peers affect me, simultaneity makes OLS hard to interpret.
- **Correlated effects:** co-workers may face the same local or sectoral shocks.
- **Sorting:** people with similar preferences, impatience, or income risk may work in the same plants.
- **Measurement error:** consumption is constructed, not directly observed.

Interpretation: The paper's empirical contribution is to design network-based instruments that address these standard threats more directly than simple peer regressions.

4.2 Why spouse links matter

The network in Figure 2 illustrates that spouses create bridges across firms. For example, the husband's co-workers may be linked to other households whose spouses work elsewhere. The paper uses shocks further away in this graph as instruments.

Interpretation: This is the key innovation. If consumption were observed for individuals but networks were isolated within firms, the instrument would be much weaker. Joint household consumption makes spouse links informative.

4.3 Robustness logic

The robustness checks ask whether the main IV coefficient survives:

- restricting to plant stayers;
- adding transition fixed effects;
- controlling for local or sector shocks;
- trimming the top and bottom of the consumption distribution;
- dropping stockholders or focusing on renters;
- changing the peer-weighting scheme;
- randomly assigning workers to firms as a placebo.

Interpretation: These exercises target the main identification threats one by one. The placebo exercise is especially important because it asks whether the estimated effect survives after the actual network is destroyed.

5 Tables and figures

5.1 Table 1 and Figure 1: data quality and sample description

Read:

- Table 1 shows a very large sample of 757,439 households.
- Mean log consumption is 12.07; mean consumption is about DKr 359k, with a median of about DKr 315k.
- The sample contains dual-worker households with substantial workplace information: tenure averages about five years, and both spouses have identifiable firm characteristics.

- Figure 1 compares tax-record consumption and survey consumption. The distributions overlap in the middle but differ in tails; the matched scatterplot shows a positive relationship but noise.

Economic message: The empirical design depends on having a very large panel because network instruments are high-dimensional and noisy. The survey validation is not perfect, but it shows that tax-record consumption captures a meaningful consumption object.

Interpretation: Table 1 is not merely descriptive. It shows why Denmark is an unusually strong laboratory: income, assets, employers, plants, spouses, and demographics can be linked. Figure 1 warns that the constructed consumption measure should be treated as spending with measurement error, motivating later robustness checks.

TABLE 1
Descriptive statistics

	Mean	Std. Dev.		Mean (Median)	Std. Dev.
Outcomes					
log Consumption	12.07	0.66	Consumption	358,873 (314,727)	324,117
log Income	12.57	0.32	Income	515,877 (484,436)	186,305
log Disposable income	12.16	0.28	Disposable income	340,041 (325,877)	117,353
			Assets	226,567 (141,792)	758,139
Socio-demographics:					
Age			Sector: Manufacturing		
Husband	42.53	9.42	Husband	25.14	
Wife	40.06	9.10	Wife	12.75	
Years of schooling			Sector: Service		
Husband	12.06	2.33	Husband	15.63	
Wife	11.70	2.33	Wife	12.22	
Occupation: Blue			Sector: Construction		
Husband	43.04		Husband	10.30	
Wife	31.63		Wife	0.99	
Occupation: White			Sector: Other		
Husband	15.83		Husband	48.93	
Wife	45.20		Wife	74.05	
Occupation: Manager			Tenure (in 1996):		
Husband	41.14		Husband	4.79	4.92
Wife	23.18		Wife	4.68	4.94
# Kids 0–6	0.38	0.66	# Kids 7–18	0.72	0.86
Workplace characteristics:					
Firm size			Type: Publicly traded		
Husband	260	65	Husband	0.46	
Wife	330	82	Wife	0.24	
Change in firm size			Type: Limited liability		
Husband	−9	32	Husband	0.08	
Wife	−13	42	Wife	0.04	
Public sector			Type: Other		
Husband	0.32		Husband	0.46	
Wife	0.61		Wife	0.72	
Number of households: 757,439					

Descriptive statistics

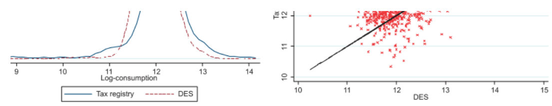

FIGURE 1
The distribution of consumption in the Tax Registry and in the Danish Expenditure Survey.

t the bottom of Table 1, we also report workplace characteristics that we use as controls nstruments. Average firm size is 260 and 330 for husbands and wives, respectively. The al employment change is negative in this period, although the standard deviation is rather As mentioned above, a larger fraction of women work in the public sector (60%) relative

Tax registry versus survey consumption

5.2 Table 2 and Figure 2: mechanisms and network identification

Read:

- Table 2 separates mechanisms: peers may affect demand functions, Euler equations, or both.
- If peer consumption enters only the Euler equation, the effect is intertemporal.
- If peer consumption enters demand functions, budget shares should move, especially for visible goods.
- Figure 2 shows the network structure used for identification. Distance-3 nodes and firm shocks provide instruments for direct peers' consumption.

Economic message: The paper is not satisfied with showing a correlation. It uses the theory to ask what kind of peer effect is present: spending/saving over time, within-period demand composition, or insurance.

Interpretation: Table 2 is the roadmap for the entire mechanism section. Figure 2 is the roadmap for identification. Together, they show why the paper's contribution is both theoretical and empirical: the authors need a model to know what to test and a network structure to identify the endogenous effect.

s for intertemporal allocation, they are also distorted, as the Euler equation is now:

$$\Delta \ln \frac{C_{it+1}}{a_{t+1}(.)} \cong \rho^{-1}\left((r-\phi)+\kappa\Delta X_{it}-\Delta \ln b_{t+1}(.)+\lambda\Delta \overline{\frac{\ln C_{it+1}}{a_{t+1}(.)}}\right)+\varepsilon_{it+1} \qquad (7)$$

Theoretical cases

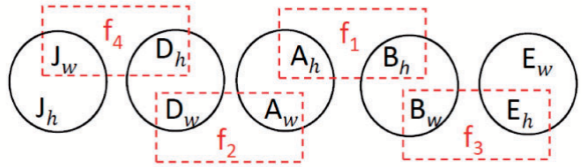

FIGURE 2
A simple example of network identification.

Network identification

5.3 Table 3 and Table 4: network scale and baseline peer effects

Read:

- Table 3 shows that husbands have about 73 weighted distance-1 peers, while wives have about 95.
- Distance-3 networks become very large: roughly 12,535 for husbands and 14,871 for wives.
- Table 4 reports the baseline estimates. OLS coefficients are about 0.11–0.13. IV estimates are larger.
- In the preferred firm-IV specification, wife peer effects are about 0.30 and husband peer effects about 0.37.
- The pooled average-peer IV coefficient is about 0.33.

Economic message: Peer effects are economically important at the average-peer level but must be interpreted carefully because an average peer-consumption shock is a shock to many co-workers at once.

Interpretation: Table 3 helps translate elasticities into economic magnitudes. Table 4 shows why IV matters: OLS understates the preferred effect, likely because measurement error and network averaging attenuate coefficients. The first-stage statistics are strong, making the IV estimates credible in this setting.

TABLE 3
Weighted network statistics (quadratic weights)

	Co-workers' group	
	Peers	Std. dev.
Distance 1		
Husband	73.30	179.77
Wife	95.07	233.34
Distance 2		
Husband	89.27	212.77
Wife	118.78	285.88
Distance 3		
Husband	12,535	33,284
Wife	14,871	40,535
	Co-workers' averages	
Variable	Wife	Husband
Age	38.24	38.59
	(5.61)	(5.62)
Years of schooling	11.69	11.81
	(1.86)	(1.82)
Share of females	0.71	0.27
	(0.26)	(0.26)
Share of blue collars	0.36	0.47
	(0.37)	(0.41)
Share of white collars	0.39	0.22
	(0.35)	(0.28)
Share of managers	0.25	0.31
	(0.34)	(0.36)
# Kids 0–6	0.28	0.27
	(0.21)	(0.21)
# Kids 7–18	0.50	0.47
	(0.30)	(0.29)

	Distance-3 peers' averages	
Variable	Wife's peers	Husband's peers
Age	38.62	38.46
	(2.48)	(2.51)
Years of schooling	11.89	11.67
	(1.26)	(1.29)
Share of females	0.45	0.62
	(0.21)	(0.20)
Share of blue collars	0.42	0.40
	(0.27)	(0.27)
Share of white collars	0.30	0.35
	(0.19)	(0.21)
Share of managers	0.29	0.25
	(0.23)	(0.22)
# Kids 0–6	0.28	0.28
	(0.09)	(0.09)
# Kids 7–18	0.46	0.47
	(0.13)	(0.14)
Firm size (in 1,000)	1.45	1.56
	(1.13)	(1.39)
Firm size growth	0.001	0.001
	(0.002)	(0.002)
Public sector	0.53	0.63
	(0.30)	(0.30)
Publicly traded	0.36	0.29
	(0.28)	(0.27)
Limited liability	0.02	0.01
	(0.08)	(0.08)
Other	0.62	0.70
	(0.29)	(0.27)

Weighted network statistics

TABLE 4
Baseline results

Model	(1) OLS FD	(2) IV FD	(3) IV FD	(4) OLS FD	(5) IV FD	(6) IV FD
Wife's peers ln C	0.11***	0.26**	0.30*	–	–	–
	(0.002)	(0.113)	(0.163)	–	–	–
Husband's peers ln C	0.13***	0.38***	0.37**	–	–	–
	(0.003)	(0.111)	(0.182)	–	–	–
Avg. peer's ln C	–	–	–	0.12***	0.32***	0.33***
	–	–	–	(0.002)	(0.061)	(0.078)
Demographic IV's	–	YES	NO	–	YES	NO
Firm IV's	–	YES	YES	–	YES	YES
p-value test $\theta_w=\theta_h$	0.000	0.510	0.819	–	–	–
p-value exogeneity test	–	0.002	0.015	–	0.001	0.004
p-value OID test	–	0.001	0.216	–	0.001	0.280
F-stat first stages						
Wife	–	44.14	78.27	–	–	–
Husband	–	43.14	59.85	–	–	–
All	–	–	–	–	62.84	111.40
Number of obs.	2,671,889	2,671,889	2,671,889	2,671,889	2,671,889	2,671,889

Notes: *,**,***= significant at 10%, 5%, 1%. Standard errors are double clustered for husbands and wives at the workplace, occupation, and education level. Dependent variable: Log of adult equivalent consumption. Individual controls (separately for husband and wife): Age, Age sq., Years of schooling, Occupation dummies, Industry dummies, Public sector dummy, Firm size, Firm growth, Firm type dummies. Household controls: Region dummies, # kids 0–6, # kids 7–18. Contextual controls (peer variables for husband and wife): Age, Age sq., Years of schooling, # kids0–6, # kids 7–18, share of female peers, shares of peers by occupation. Demographic IV's: Age, Age sq., Years of schooling, # kids0-6, # kids 7-18, share of

Baseline network-effect estimates

5.4 Figure 3 and Table 5: heterogeneity and correlated effects

Read:

- Figure 3 shows that network effects are fairly stable across network size.
- Effects appear stronger among lower-education workers, in male-dominated peer groups, during periods of growth, and among workers with some tenure.

- Table 5 shows that the average-peer effect remains positive across plant stayers, transition fixed effects, local shocks, and sector shocks.
- The coefficient ranges from about 0.29 to 0.44, with strong first stages.

Economic message: The heterogeneity is consistent with social learning or social comparison requiring time and interaction. The robustness table weakens the concern that peer effects are simply shared shocks.

Interpretation: The tenure result is particularly revealing: new employees show little effect, while longer-tenure workers show large peer effects. This suggests that the effect is not just mechanical income comovement; it may require knowing peers' habits, norms, or consumption aspirations.

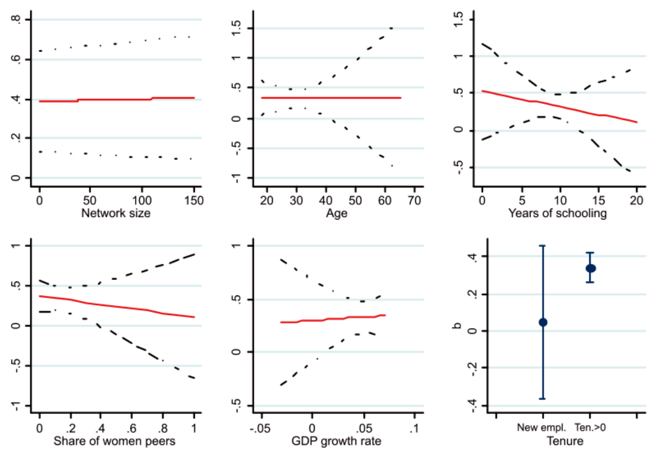

FIGURE 3
Heterogeneous network effects.

Heterogeneous network effects

TABLE 5
Robustness to correlated effects

Model	(1) Baseline	(2) Stayers	(3) Transition FE	(4) Local shocks	(5) Sector shocks
Avg. peer's ln C	0.33***	0.31***	0.29***	0.44***	0.33***
	(0.078)	(0.106)	(0.063)	(0.049)	(0.073)
F-stat first stage	111.40	59.09	58.91	220.9	127.7
Number of obs.	2,671,889	1,323,086	2,671,889	2,671,889	2,671,889

Notes: *,**,***= significant at 10%, 5%, 1%. Standard errors are double clustered for husbands and wives at the workplace, occupation, and education level. Dependent variable: Log of adult equivalent consumption. Individual controls (separately for husband and wife): Age, Age sq., Years of schooling, Occupation dummies, Industry dummies, Public sector dummy, Firm size, Firm growth, Firm type dummies. Household controls: Region dummies, # kids 0–6, and # kids 7–18. Contextual controls (peer variables for husband and wife): Age, Age sq., Years of schooling, # kids 0–6, # kids 7–18, share of female peers, and shares of peers by occupation. Firm IV's: Public sector dummy, Firm size, Firm growth, and Firm type dummy. We also control for year fixed effects. The regression in column (1) is the baseline (from Table 4, column 6). For details on the other specifications, see the main text.

Robustness to correlated effects

5.5 Tables 6 and 7: measurement, weighting, and the saving channel

Read:

- Table 6 shows that the baseline effect is robust to trimming, dropping stockholders, using renters, and alternative weighting schemes.
- The placebo network effect is 0.01 and statistically insignificant, suggesting the true network assignment matters.
- Table 7 decomposes peer consumption into peer income and peer saving behavior.
- Peer income has a small and imprecise coefficient around 0.09, while the peer saving-rate component is about 0.33 and significant.

Economic message: The consumption response appears more related to peers' spending/saving decisions than to peer labor-supply or income changes. That supports an intertemporal consumption-channel interpretation.

Interpretation: Table 6 is a credibility table: the finding is not driven by tails, investors, renters, or a specific weighting formula. Table 7 is a mechanism table: households seem to react to how peers allocate income between consumption and saving, not merely to whether peers work more or earn more.

TABLE 6

Robustness to measurement error and weighting scheme

Model	(1) Baseline	(2) 1% trim	(3) Drop stockh.	(4) Renters	(5) Linear w.	(6) Exp. quadr.	(7) Sharp	(8) Placebo
Avg. peer's ln C	0.33***	0.32***	0.36***	0.24***	0.62***	0.30***	0.44***	0.01
	(0.078)	(0.066)	(0.088)	(0.082)	(0.164)	(0.082)	(0.083)	(0.028)
F-stat first stage	111.40	101.90	82.38	34.69	127.03	77.67	88.57	–
Number of obs.	2,671,889	2,588,299	2,016,137	318,317	2,671,889	2,671,823	2,171,426	2,671,889

Notes: *,**,***= significant at 10%, 5%, 1%. Standard errors are double clustered for husbands and wives at the workplace, occupation, and education level. Dependent variable: Log of adult equivalent consumption. Individual controls (separately for husband and wife): Age, Age sq., Years of schooling, Occupation dummies, Industry dummies, Public sector dummy, Firm size, Firm growth, and Firm type dummies. Household controls: Region dummies, # kids 0–6, and # kids 7–18. Contextual controls (peer variables for husband and wife): Age, Age sq., Years of schooling, # kids 0–6, # kids 7–18, share of female peers, and shares of peers by occupation. Firm IV's: Public sector dummy, Firm size, Firm growth, and Firm type dummy. We also control for year fixed effects. The regression in column (1) is the baseline (from Table 4, column 6). For details on the other specifications, see the main text.

Measurement and weighting robustness

TABLE 7

Response to effort versus saving rate

Avg. peer's ln Y	0.09
	(0.166)
Avg. peer's saving rate	0.33***
	(0.078)
Number of obs.	2,671,889

Notes: *,**,***= significant at 10%, 5%, 1%. Standard errors are double clustered for husbands and wives at the workplace, occupation, and education level. Dependent variable: Log of adult equivalent consumption. Individual controls (separately for husband and wife): Age, Age sq., Years of schooling, Occupation dummies, Industry dummies, Public sector dummy, Firm size, Firm growth, and Firm type dummies. Household controls: Region dummies, # kids 0–6, and # kids 7–18. Contextual controls (peer variables for husband and wife): Age, Age sq., Years of schooling, # kids 0–6, # kids 7–18, share of female peers, and shares of peers by occupation. Firm IV's: Public sector dummy, Firm size, Firm growth, and Firm type dummy. We also control for

Effort versus saving rate

5.6 Table 8 and Figure 4: demand estimation and visible goods

Read:

- Table 8 estimates budget-share responses for visible and not-visible consumption categories.
- Own consumption shifts budget shares: visible goods rise with own consumption and not-visible goods fall.
- Peer consumption has very small, statistically insignificant effects on budget shares.
- Figure 4 plots good-level estimates against the Heffetz visibility index. There is no clean positive relationship between peer effects and visibility.

Economic message: The evidence does not support a strong conspicuous-consumption channel in which households mimic peers by moving expenditure toward visible goods.

Interpretation: This is one of the paper's most important negative results. Peer consumption affects total consumption, but it does not strongly reallocate spending toward observable categories. That pushes the interpretation away from pure status competition and toward intertemporal spending behavior.

TABLE 8

Demand estimation

	ALL sample		MATCH sample	
	Visible cons. (1)	Not-visible cons. (2)	Visible cons. (5)	Not-visible cons. (6)
ln C	0.205***	−0.132***	0.270***	−0.165***
	(0.007)	(0.007)	(0.018)	(0.017)
Avg. peer's ln C	0.005	−0.000	0.004	0.007
	(0.011)	(0.011)	(0.028)	(0.027)
Observations	2,435	2,437	454	452

Note: *,**,***= significant at 10%, 5%, 1%. The dependent variables are budget shares for three consumption groups: Visible, Not-visible and Neutral. The omitted category is Neutral. Individual controls: Age, Age sq., Years of schooling, Occupation dummies, Industry dummies, Public sector dummy, Firm size, Firm growth, and Firm type dummies. Household controls: Year dummies, Region dummies, # kids 0–6, and # kids 7–18. Columns (1)–(4) are OLS estimates;

Demand estimation

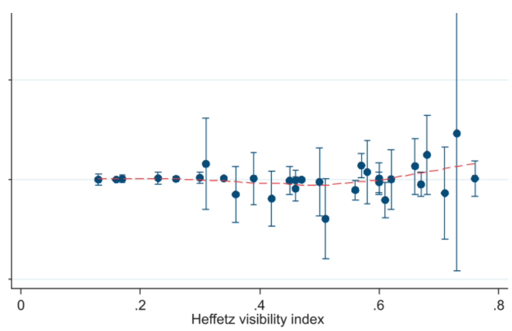

FIGURE 4

ıship between the shift in budget shares due to peer consumption and the Heffetz visibility index.

ere were any intratemporal effects. However, the disaggregated evidence is similar ted above. The effect of peer consumption on the budget share on good *j* appears of the degree of conspicuousness of the good. The relationship is increasing only for

Visibility index and budget-share shifts

5.7 Tables 9 and 10: risk sharing and aggregate policy simulations

Read:

- Table 9 tests whether survey-versus-tax-record consumption differences behave like informal risk sharing.
- Risk sharing would predict a negative coefficient on own income growth and a positive coefficient on peer income growth.

- The own-income coefficient is positive, not negative, and peer income is not significant.
- Table 10 simulates transfer policies. Transfers to the bottom 10% have the largest implied multiplier, 1.034.
- A balanced-budget transfer from the top to the bottom yields a multiplier of 1.021 and the largest compression in consumption dispersion.

Economic message: The policy multiplier depends on who receives the transfer and how connected that group is. Sparse networks make the simple scalar social multiplier misleading.

Interpretation: Table 9 closes off the insurance interpretation. Table 10 shows why network effects matter for macro policy: targeting low-consumption, highly connected groups can both increase aggregate consumption and compress inequality. This is a distributional multiplier, not just an aggregate multiplier.

TABLE 9
Tests of risk sharing

	ALL sample	
	(1)	(2)
$\Delta \ln Y$	0.135*	0.131*
	(0.069)	(0.069)
$\overline{\Delta \ln Y}$		0.046
		(0.078)
Observations	2,454	2,454

Note: *,**,***= significant at 10%, 5%, 1%. Individual controls: Age, Age sq., Years of schooling, Occupation dummies, Industry dummies, Public sector dummy, Firm size, Firm growth, and Firm type dummies. Household controls: Year dummies, Region dummies, # kids 0–6, and # kids 7–18. The dependent variable is the difference between log consumption in the survey and log consumption in the tax records.

Risk-sharing tests

TABLE 10
Counterfactual policy simulations

Transfer recipients	Implied multiplier	Std.Dev. log cons.	90-10 log difference	50-10 log difference	90-50 log difference
Baseline	–	0.729	1.5795	0.8630	0.7165
Top 10%	1.012	0.736	1.5818	0.8631	0.7187
Random 10%	1.017	0.718	1.5608	0.8506	0.7102
Bottom 10%	1.034	0.601	1.4525	0.7349	0.7176
Balanced budget	1.021	0.593	1.4216	0.7347	0.6869

Counterfactual policy simulations

6 Short summary

- The paper studies whether co-worker consumption affects household consumption.
- It uses Danish administrative data from 1980–1996 and constructs consumption from income and asset changes.
- Peer groups are workplace networks based on plant, occupation, and education.
- Spousal links let the authors build non-overlapping network instruments.
- The main IV estimates imply an average peer-consumption elasticity of about 0.3.
- The effect is robust to correlated effects, local shocks, sector shocks, trimming, alternative weighting schemes, and placebo networks.
- Heterogeneity suggests effects are stronger among longer-tenure workers and in settings where co-worker interaction is more meaningful.
- Mechanism tests find little evidence of budget-share distortions toward visible goods.
- Risk-sharing tests also do not support informal insurance as the main explanation.
- The paper's preferred interpretation is that peer effects affect intertemporal consumption decisions: households adjust spending/saving in response to peers.
- Policy simulations show that network structure changes fiscal multipliers and distributional effects.

7 Conclusion

7.1 Core conclusion

De Giorgi, Frederiksen, and Pistaferri show that peer consumption matters. Using rich Danish administrative data and workplace networks, they estimate economically meaningful consumption network effects. The preferred IV coefficient around 0.3 implies that households adjust consumption when their co-worker peer groups adjust consumption.

7.2 Mechanism conclusion

The paper's strongest mechanism evidence points to intertemporal consumption effects. Households do not appear to simply buy more visible goods when peers consume more, and the demand estimates do not show a strong visibility gradient. Nor does the survey-versus-tax-record test support informal risk sharing. The effect is therefore closer to a "keeping-up" effect in total spending or saving than to a conspicuous-consumption reshuffling across goods or a co-worker insurance arrangement.

7.3 Identification conclusion

The identification strategy is valuable because it uses the actual network geometry of households and workplaces. Distance-3 nodes and firm-level shocks help isolate variation in peer consumption that is less contaminated by direct common shocks. The key assumption is that these distant shocks affect the household only through direct peers. This is plausible but not mechanical; it depends on adequately controlling for aggregate, sector, local, and firm-level shocks.

7.4 Policy implication

The policy simulations show that network structure affects stimulus design. Transfers to the bottom of the consumption distribution produce larger multipliers than transfers to the top because low-consumption households are more connected to other low-consumption households and have stronger propagation effects. A balanced-budget redistribution from high to low consumption households both raises aggregate consumption and compresses the distribution.

7.5 Welfare implication

The welfare interpretation depends on the mechanism. If peer effects lead households to over-borrow or under-save, they may create intertemporal distortions. If they reflect preferences over relative consumption, the welfare benchmark is more subtle. If they reflected risk sharing, the welfare implications would be positive, but the paper finds little evidence for that interpretation.

7.6 Broader takeaway

The paper changes how we should think about consumption spillovers. Peer effects are not only neighborhood or family phenomena. Workplace networks can transmit consumption behavior, and household links can connect distinct labor-market networks. The resulting propagation matters for both micro consumption decisions and macro policy multipliers.

7.7 Limitations and future research

Several questions remain open. First, the paper observes co-worker networks but not friendship intensity, conversations, or exposure to peer consumption. Second, the consumption measure is inferred from tax records and may be noisy. Third, Denmark's welfare state and high female labor-force participation may affect external validity. Future work could compare workplace networks with family, neighborhood, school, or online networks; measure peer exposure directly; and study whether network effects differ in economies with weaker social insurance and more credit constraints.